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Acknowledgment

We would like to express our sincere gratitude to **Beau Roi Pvt Ltd** and mentors for providing us with the opportunity to work on the **AI Resume Intelligence and Recruitment Analytics Platform** during our internship period. This project provided us with practical exposure to real-time recruitment automation systems, dashboard analytics, database integration, and intelligent data management.

We sincerely thank our project mentors and guides for their continuous support, technical guidance, and valuable suggestions throughout the development process. Their encouragement helped us successfully complete the project and improve our technical and analytical skills.

We also thank our teammates for their cooperation, teamwork, and dedication during every stage of the project development. This internship experience greatly improved our understanding of Python programming, MySQL database management, Tableau analytics, dashboard visualization, and software development practices.

About the Company

Company Profile

Beau Roi Enterprise Group is a forward-thinking technology company specializing in innovative software solutions. The company focuses on delivering high-quality digital products that meet the evolving needs of modern businesses. With a team of experienced developers and designers, **Beau Roi** maintains a strong commitment to excellence, innovation, and customer satisfaction.

The company culture strongly promotes continuous learning, teamwork, and the use of modern technologies to address complex business challenges effectively. During our internship from March 17 to May 17, we had the opportunity to work on real-time projects under the guidance of experienced professionals who shared valuable industry knowledge and best development practices.

Internship Experience

Our internship at Beau Roi Pvt Ltd spanned two months from March 17 to May 17, during which we worked as a team comprising Muthu Kumaran Siva T and Ahamed Sherif on the AI Resume Intelligence and Recruitment Analytics Platform project. The internship experience was divided into multiple phases, with each phase focusing on different aspects of software development, database integration, automation systems, and dashboard analytics.

Phase 1: Requirement Analysis and Project Planning

During the initial phase, we analyzed the limitations of traditional recruitment systems and identified the need for an intelligent recruitment analytics platform. We conducted discussions regarding system requirements, recruitment workflows, dashboard planning, and database structure design.

We finalized the major project modules including:

- Candidate Management System
- Resume Evaluation System
- Tableau Dashboard Analytics
- MySQL Database Integration
- QR-Based Candidate Registration
- PDF Report Generation

This phase helped us understand real-world recruitment challenges and the importance of automation in modern hiring systems.

Phase 2: Frontend Development

In the second phase, we focused on developing the frontend interface using Python Tkinter. We designed an interactive desktop application that allows recruiters to manage candidate information efficiently.

The frontend development included:

- Candidate profile forms
- Search and sorting functionalities
- Candidate table management
- Dashboard control sections
- AI evaluation panels
- Interactive user interface components

We ensured that the UI was user-friendly, responsive, and capable of handling recruitment operations smoothly.

Phase 3: Backend Development and Database Integration

During this phase, we implemented backend functionalities using Python. The backend handled candidate data processing, resume evaluation, CSV synchronization, PDF generation, and dashboard communication.

We integrated MySQL database support for structured recruitment data storage. Automatic table creation and data synchronization functionalities were implemented successfully.

- The backend system managed:
 - Candidate insertion and updates
 - Resume score generation
 - Candidate filtering and analytics
 - CSV and Excel synchronization
 - Database operations

This phase improved our understanding of backend logic implementation and database management systems.

Phase 4: Dashboard Analytics and Visualization

The final phase focused on analytical dashboard development and visualization. We integrated Tableau dashboards for professional recruitment analytics and interactive reporting.

The dashboard included:

- Total candidate analysis

- Department-wise candidate distribution
- Role-based analytics
- Salary and experience comparison
- Candidate status visualization
- Skill analysis charts
- Recruitment KPI monitoring

We also developed Python-based analytical charts using Matplotlib for additional visualization support.

This phase helped us understand business intelligence concepts, dashboard design principles, and data visualization techniques.

Introduction

The AI Resume Intelligence and Recruitment Analytics System is an intelligent recruitment management platform developed to automate candidate screening, resume analysis, recruitment analytics, and reporting processes. In modern organizations, handling large numbers of resumes manually becomes difficult and time-consuming. Traditional recruitment systems often depend on spreadsheets and manual shortlisting methods, which reduce efficiency and increase workload for HR departments.

This project was developed to simplify and modernize the recruitment process using Python automation, machine learning, database integration, and dashboard analytics. The system integrates technologies such as Tkinter, Flask, MySQL, Tableau, Matplotlib, and Scikit-learn to provide a centralized recruitment management environment.

The platform allows recruiters to manage candidate details, upload resumes, automatically evaluate candidate profiles, generate AI-based scores, create professional PDF reports, and monitor recruitment analytics through interactive dashboards. The system also supports QR-based candidate registration, enabling fast and efficient mobile-based data entry.

The project provides an efficient and scalable solution for intelligent recruitment management by combining automation, analytics, and business intelligence visualization into one integrated system.

Objective

The main objective of the AI Resume Intelligence and Recruitment Analytics System is to automate and improve the recruitment process using intelligent technologies and analytical visualization tools. The system aims to reduce manual workload, improve recruitment accuracy, and provide better decision-making support through interactive dashboards and AI-based evaluation.

The objectives of the project are:

- To automate candidate data management
- To reduce manual resume screening effort
- To implement AI-based resume evaluation and scoring
- To create centralized recruitment data storage using MySQL
- To provide real-time recruitment analytics and visualization
- To generate professional PDF reports automatically
- To implement QR-based candidate registration
- To support importing and exporting recruitment datasets
- To improve recruitment efficiency and hiring decisions
- To integrate Tableau dashboards for business intelligence reporting

The project also aims to provide practical exposure to real-world software development, dashboard analytics, database integration, and intelligent automation systems.

Problem Statement

In many organizations, recruitment processes are still performed manually using spreadsheets, paper-based records, and disconnected reporting systems. Managing large numbers of candidate resumes becomes difficult, time-consuming, and inefficient. HR teams often spend significant time filtering resumes, organizing candidate details, generating reports, and performing candidate shortlisting.

Traditional recruitment systems face several limitations such as:

- Manual resume verification and screening
- Difficulty in managing large candidate datasets

- Lack of centralized recruitment dashboards
- No intelligent candidate scoring mechanism
- Poor data visualization and reporting •
- Time-consuming shortlisting process •
- Lack of real-time analytical insights
- Data duplication and inconsistency
- Difficulty in recruitment tracking and monitoring

Most existing systems do not provide advanced analytical dashboards or AI-based evaluation systems for understanding recruitment trends and candidate performance. This creates challenges in decision-making and increases recruitment processing time.

To overcome these issues, the AI Resume Intelligence and Recruitment Analytics System was developed. The proposed system automates recruitment management, resume evaluation, dashboard analytics, PDF reporting, and database synchronization using intelligent technologies and visualization tools.

Literature Survey

Recruitment automation systems have become increasingly important in modern organizations due to the growing number of job applications and the need for efficient hiring processes. Several existing recruitment systems focus on candidate management and basic database operations, but many lack intelligent analytics, dashboard integration, and AI-based resume evaluation.

Traditional recruitment management systems mainly support:

- Resume storage
- Candidate record maintenance
- Basic shortlisting
- Spreadsheet-based analytics

However, these systems often require significant manual effort and provide limited analytical capabilities.

Recent advancements in artificial intelligence and machine learning have introduced automated resume screening systems capable of analyzing candidate skills, experience, and educational qualifications. Many organizations are also adopting business intelligence tools such as Tableau for interactive dashboard analytics and recruitment visualization.

Several research studies highlight the importance of:

- AI-based resume parsing
- Machine learning candidate evaluation
- Recruitment dashboard analytics
- Automated reporting systems
- QR-based registration systems

The proposed project combines multiple modern technologies including:

- Python automation
- Machine learning
- Tableau visualization
- QR integration
- Database management
- PDF report generation

into a single intelligent recruitment analytics platform. This integration improves recruitment efficiency, reduces manual workload, and provides better hiring insights through interactive analytical dashboards.

Existing System

The traditional recruitment process mainly depends on manual resume screening, spreadsheet management, and disconnected reporting methods. HR departments usually maintain candidate records using Excel sheets or paper-based documentation, making recruitment operations difficult and time-consuming.

In the existing system:

- Candidate details are entered manually

- Resume verification is performed manually
- Recruitment analytics are limited
- Reports are created manually
- Candidate filtering requires more effort
- Dashboard visualization is not available
- Centralized database systems are often absent

The existing system has several disadvantages:

Limitations of Existing System

- Time-consuming recruitment process
- Manual resume shortlisting
- Difficulty in handling large candidate datasets
- Lack of automation and intelligent evaluation
- Poor analytical visualization
- No real-time recruitment insights
- Increased possibility of human error
- Limited scalability and flexibility

These limitations reduce recruitment efficiency and create difficulties in candidate management and decision-making.

Proposed System

The proposed AI Resume Intelligence and Recruitment Analytics System introduces an intelligent and automated recruitment management platform integrating AI technologies, database systems, dashboard analytics, and reporting tools.

The system automates multiple recruitment operations such as:

- Candidate data management
- Resume parsing and evaluation
- AI-based scoring
- Recruitment analytics
- PDF report generation

- QR-based candidate registration

The proposed system contains:

- Interactive Tkinter dashboard
- AI evaluation system
- MySQL database integration
- Tableau analytics dashboard
- Matplotlib graphical visualization
- QR code-based mobile entry
- CSV and Excel synchronization

The platform allows recruiters to:

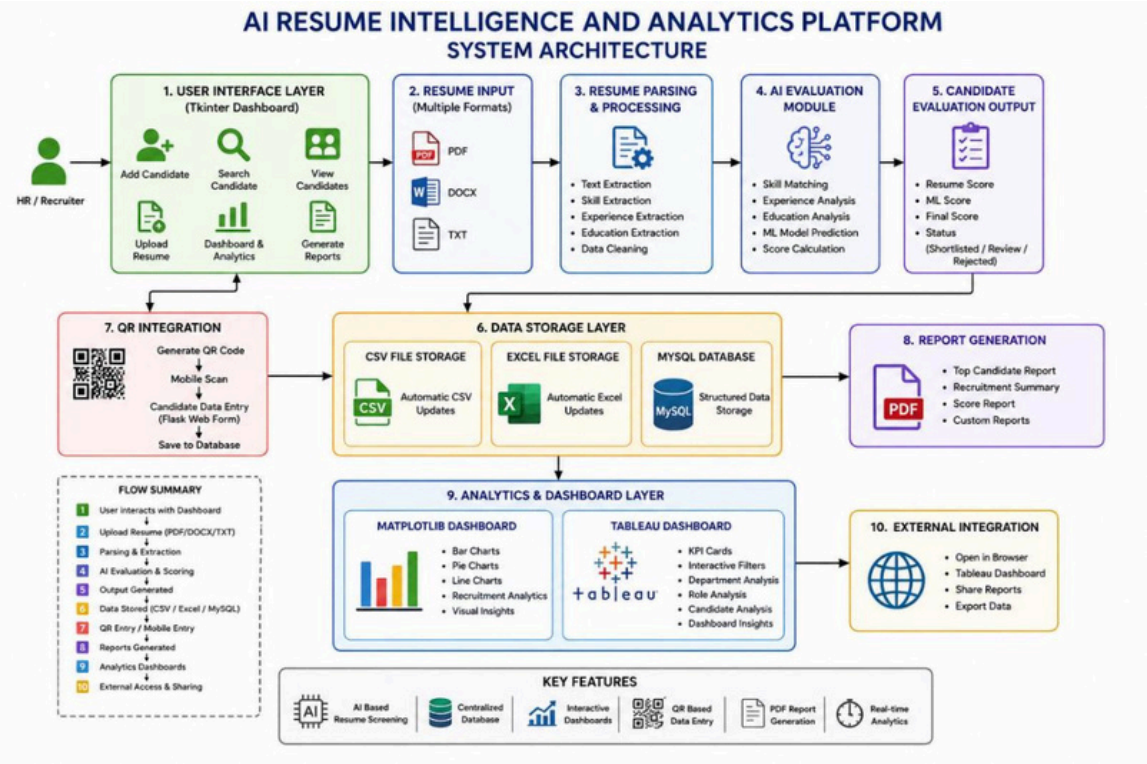
- Add and manage candidate profiles
- Upload resumes in multiple formats
- Automatically evaluate candidate skills
- Generate recruitment analytics
- Monitor candidate performance
- Create professional reports

Advantages of Proposed System

- Reduced manual workload
- Faster recruitment processing
- Intelligent candidate evaluation
- Centralized recruitment management
- Interactive dashboard analytics
- Real-time recruitment monitoring
- Automated report generation

The proposed system provides a scalable and efficient recruitment management. solution for modern intelligent

System Architecture



Technologies Used

The project uses multiple technologies and libraries for implementing recruitment automation, analytics visualization, database management, and AI-based evaluation.

Technology	Purpose
Python	Core programming language
Tkinter	GUI dashboard development
Flask	QR-based candidate registration
MySQL	Structured database storage

Technology	Purpose
Pandas	Data handling and CSV processing
Matplotlib	Recruitment analytics charts
Tableau	Interactive business intelligence dashboards
Scikit-learn	Machine learning and AI scoring
pdfplumber	PDF resume parsing
python-docx	Word resume reading
reportlab	PDF report generation
OpenPyXL	Excel file handling
CSV Module	CSV synchronization
joblib / pickle	ML model storage
qrcode	QR code generation

These technologies together helped in building a complete intelligent recruitment analytics platform.

Methodology

The project follows a step-by-step methodology for managing recruitment operations and generating analytical insights.

Step 1: Candidate Data Entry

Candidate information is entered through:

- Main dashboard
- QR-based mobile entry
- CSV import system

The entered data includes:

- Name
- Role
- Department
- Experience
- Education
- Skills
- Salary

Step 2: Resume Upload and Parsing

Candidates' resumes are uploaded in:

- PDF format
- DOCX format
- TXT format

The system automatically extracts:

- Skills
- Educational qualifications
- Experience
- Technical keywords

using resume parsing libraries.

Step 3: AI-Based Resume Evaluation

The AI module evaluates candidate profiles using:

- Skill matching
- Experience analysis
- Education analysis
- Machine learning prediction

The system generates:

- Resume Score
- ML Score
- Final Recruitment Score

Candidates are categorized as:

- Shortlisted
- Review
- Rejected

Step 4: Database Storage

Candidate data is automatically stored in:

- CSV files
- Excel sheets
- MySQL database

This provides centralized recruitment data management.

Step 5: Dashboard Analytics

The system generates recruitment analytics using:

- Matplotlib charts
- Tableau dashboards

The dashboards display:

- Candidate distribution
- Department analysis
- Salary comparison
- Recruitment status
- KPI metrics

Step 6: PDF Report Generation

The system generates professional PDF reports containing:

- Top candidates
- Recruitment summary
- Candidate scores
- Final recruitment status

This improves recruitment documentation and HR reporting.

Implementation

The implementation phase involved developing and integrating all recruitment automation modules into a centralized intelligent recruitment platform.

Frontend Implementation

The frontend was developed using Python Tkinter. The dashboard interface contains:

- Candidate profile forms
- Search and sorting modules
- Candidate tables
- Dashboard controls
- AI evaluation panels

The UI provides easy recruitment management and real-time interaction.

Backend Implementation

The backend logic was implemented using Python automation techniques.

Backend Functionalities:

- Resume parsing
- Candidate scoring
- CSV synchronization
- Data processing
- PDF generation
- Dashboard integration

The backend controls the complete recruitment workflow.

Database Implementation

MySQL database integration was implemented for structured recruitment data storage.

Database Operations:

- Insert
- Update
- Delete
- Search
- SQL filtering

Automatic table creation and synchronization functionalities were implemented successfully.

QR Integration

Flask integration was used for QR-based candidate registration.

QR Workflow:

1. QR code generation
2. Mobile scanning
3. Candidate data transfer
4. Automatic dashboard update

This reduced manual candidate entry effort.

Dashboard Implementation

The project contains two analytical dashboard systems:

1. Matplotlib Dashboard

Provides:

- Bar charts
- Pie charts
- Recruitment analytics

- Candidate comparison graphs

2. Tableau Dashboard

Provides:

- KPI cards
- Interactive filtering
- Recruitment visualization
- Business intelligence analytics

The dashboards improve recruitment analysis and decision-making.

PDF Report Generation

The system automatically generates:

- Top candidate reports
- Recruitment summary reports
- Candidate evaluation reports

using ReportLab PDF libraries.

The implementation successfully integrated automation, analytics, AI scoring, database systems, and dashboard visualization into one intelligent recruitment platform.

Results and Output

The AI Resume Intelligence and Recruitment Analytics System was successfully developed and implemented. The project achieved all major objectives related to recruitment automation, AI-based evaluation, dashboard analytics, and report generation.

The system successfully performed the following operations:

- Candidate data management
- Resume upload and parsing
- AI-based candidate evaluation
- Recruitment score generation
- QR-based candidate registration

- CSV and MySQL synchronization
- PDF report generation
- Interactive dashboard analytics

The Tkinter dashboard provided an interactive interface for managing candidate profiles and recruitment operations efficiently. Candidate details entered through the UI were automatically synchronized with CSV files and MySQL databases.

The resume parsing module successfully extracted:

- Skills
- Education
- Experience
- Technical keywords

from uploaded PDF and DOCX resumes.

The AI evaluation system generated:

- Resume Score
- ML Score
- Final Recruitment Score

based on candidate qualifications and skill matching.

The Tableau dashboard successfully displayed:

- Total candidates
- Department-wise distribution
- Candidate status analysis
- Salary vs Experience comparison
- Role-based analytics
- Recruitment KPI visualization

The Matplotlib dashboard provided graphical visualization using bar charts and pie charts for recruitment monitoring.

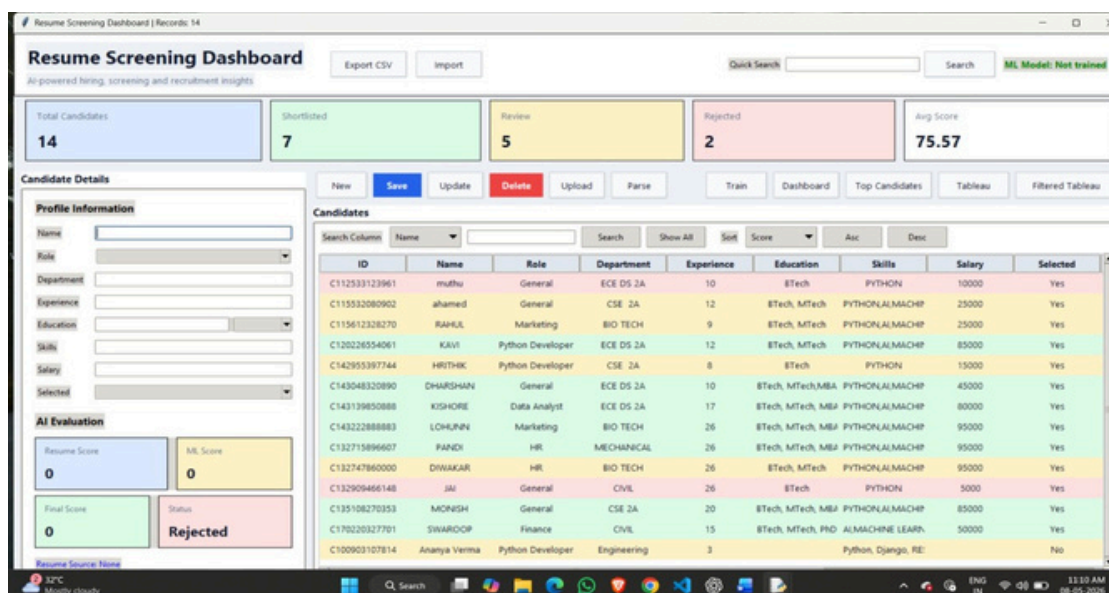
The QR-based candidate entry system enabled mobile-based registration and real-time synchronization with the dashboard and database.

The PDF generation module successfully created:

- Recruitment summary reports
- Top candidate reports
- Candidate evaluation reports

The overall system improved recruitment efficiency, reduced manual workload, and provided centralized recruitment analytics with interactive dashboard visualization.

MainDashboard

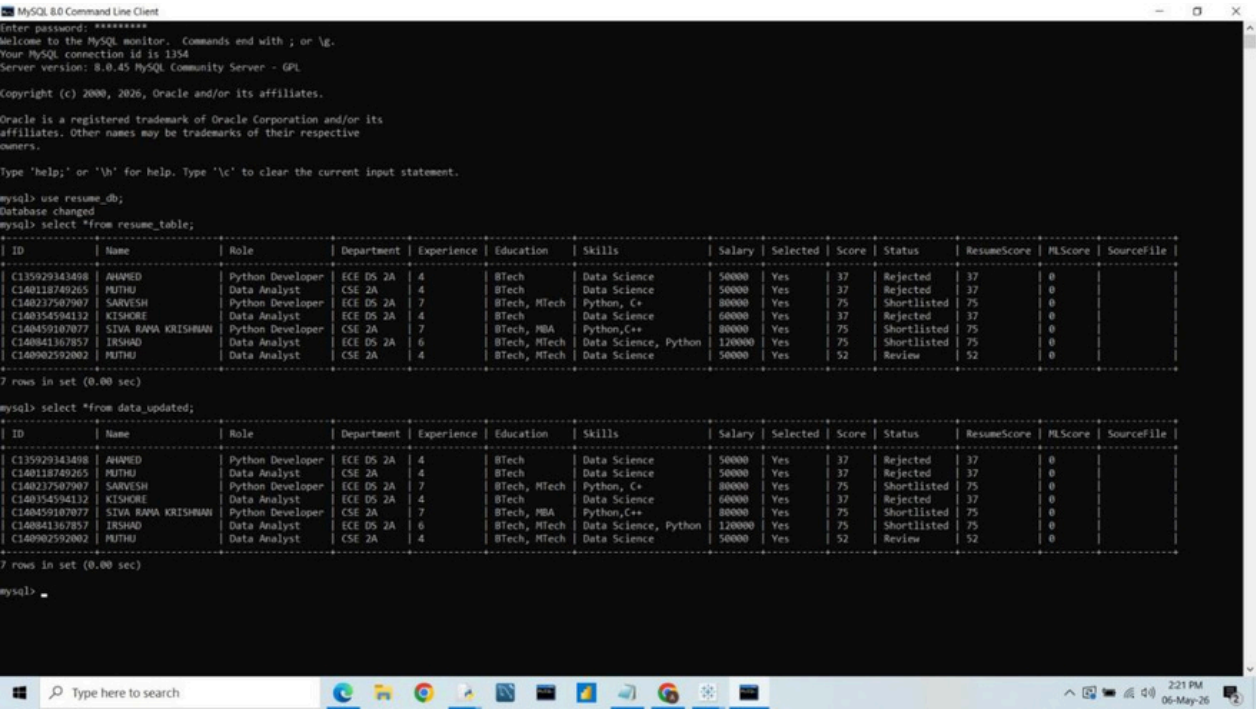


Excel Auto Entry System

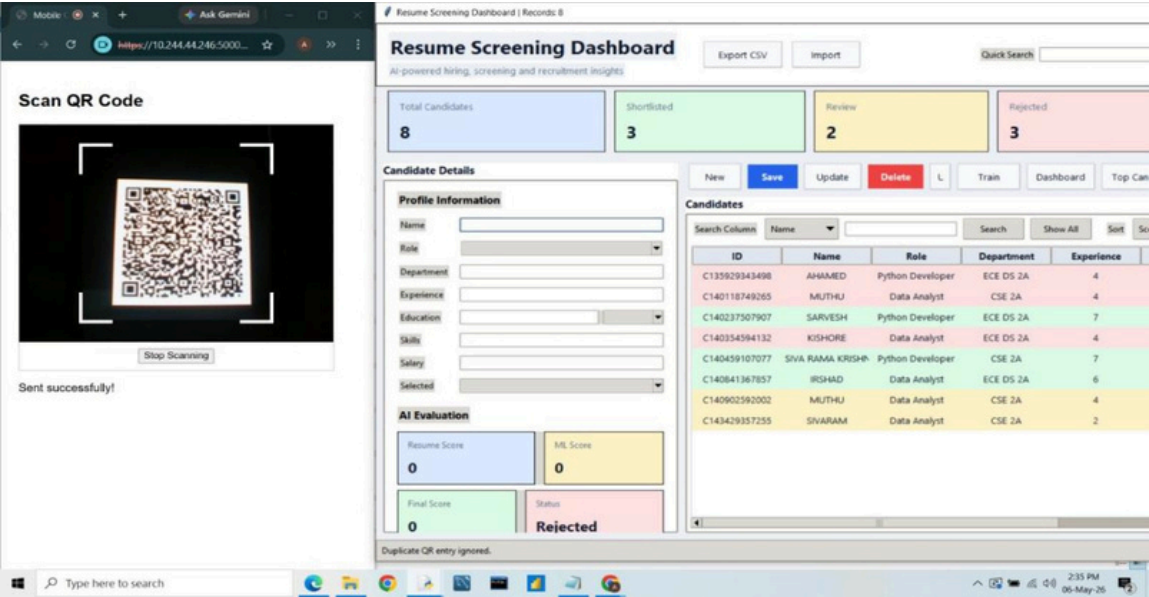
The screenshot displays the 'Excel Auto Entry System' with the following data:

ID	Name	Role	Department	Experience	Education	Skills	Salary	Selected	Status	ResumeFile	MLScore	SourceFile
C1359293	AHMED	Python Dev	ECE DS 2A	4	BTech	Data Scienc	50000	Yes	37	Rejected	37	0
C1401187	MUTHU	Data Analyst	CSE 2A	4	BTech	Data Scienc	50000	Yes	37	Rejected	37	0
C1402370	SARVESH	Python Dev	ECE DS 2A	7	BTech, MT	Python, C	80000	Yes	75	Shortlisted	75	0
C1403450	KISHORE	Data Analyst	ECE DS 2A	4	BTech	Data Scienc	60000	Yes	37	Rejected	37	0
C1404591	SIVA RAMU	Python Dev	CSE 2A	7	BTech, MT	Python, C	80000	Yes	75	Shortlisted	75	0
C1408413	SRINIKH	Data Analyst	ECE DS 2A	6	BTech, MT	Data Scienc	120000	Yes	75	Shortlisted	75	0
C1409023	MUTHU	Data Analyst	CSE 2A	4	BTech, MT	Data Scienc	50000	Yes	52	Review	52	0

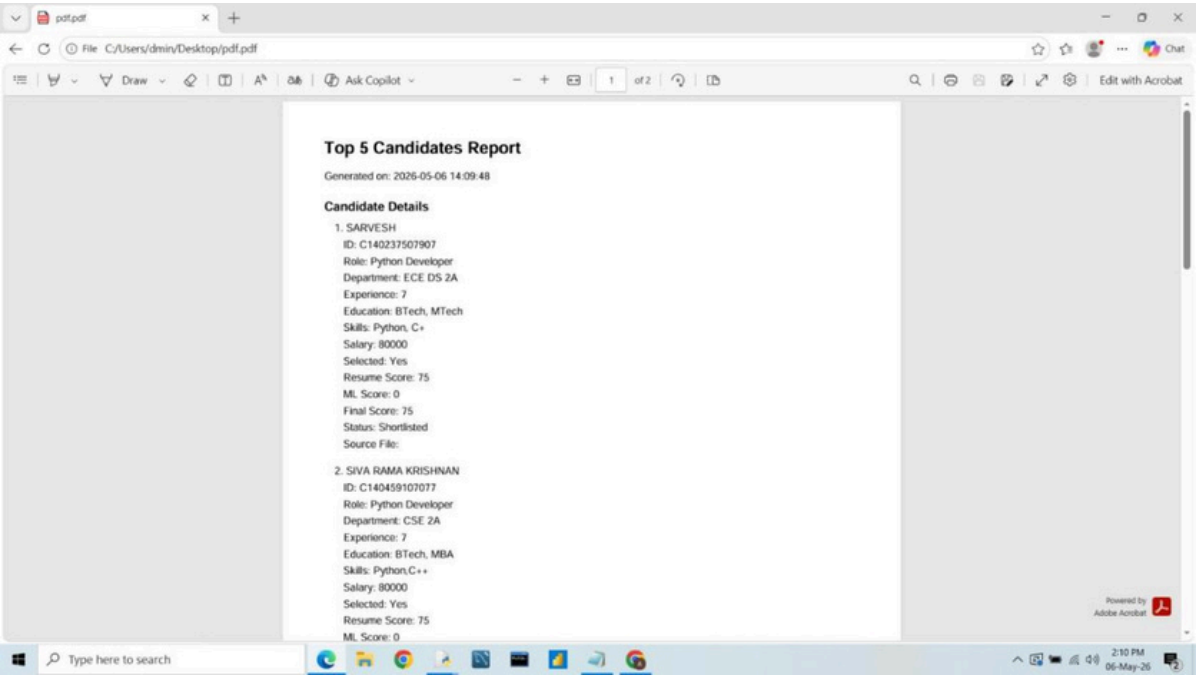
MySQL Database Integration



QR Code Entry System



PDF Report Generation



Matplotlib Dashboard

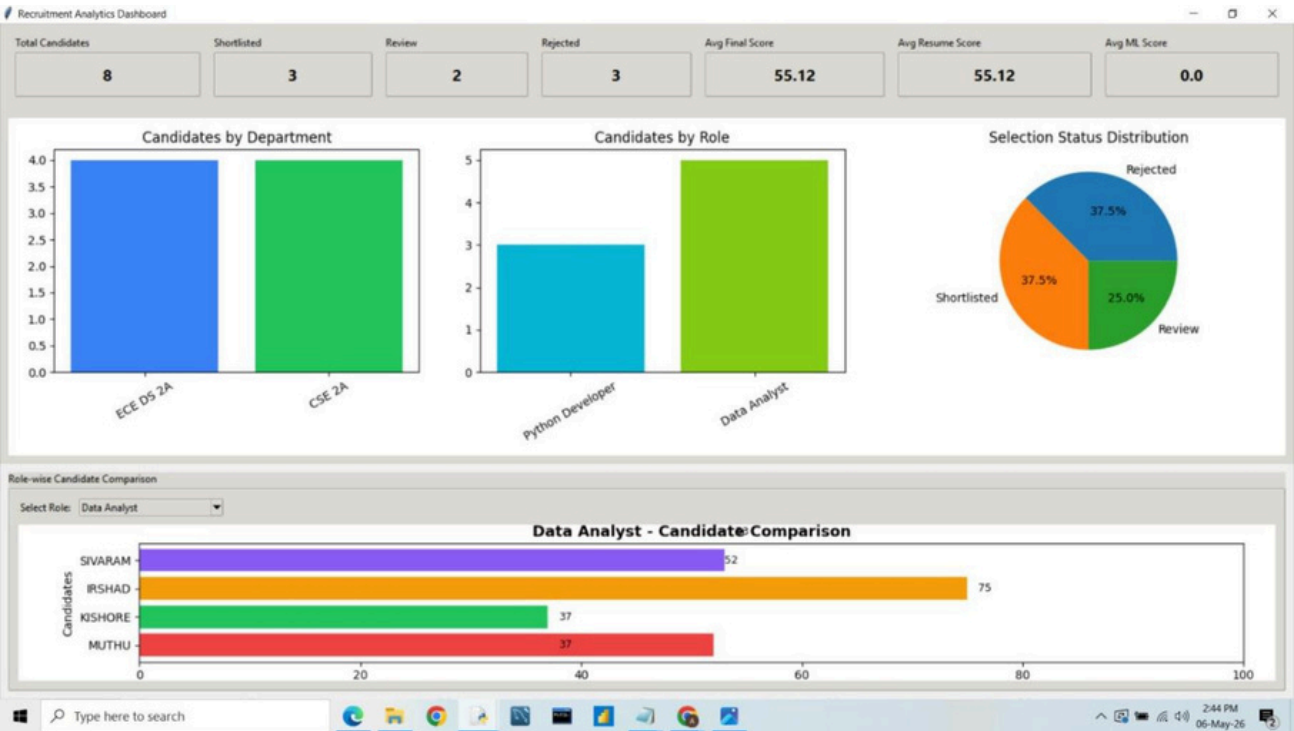
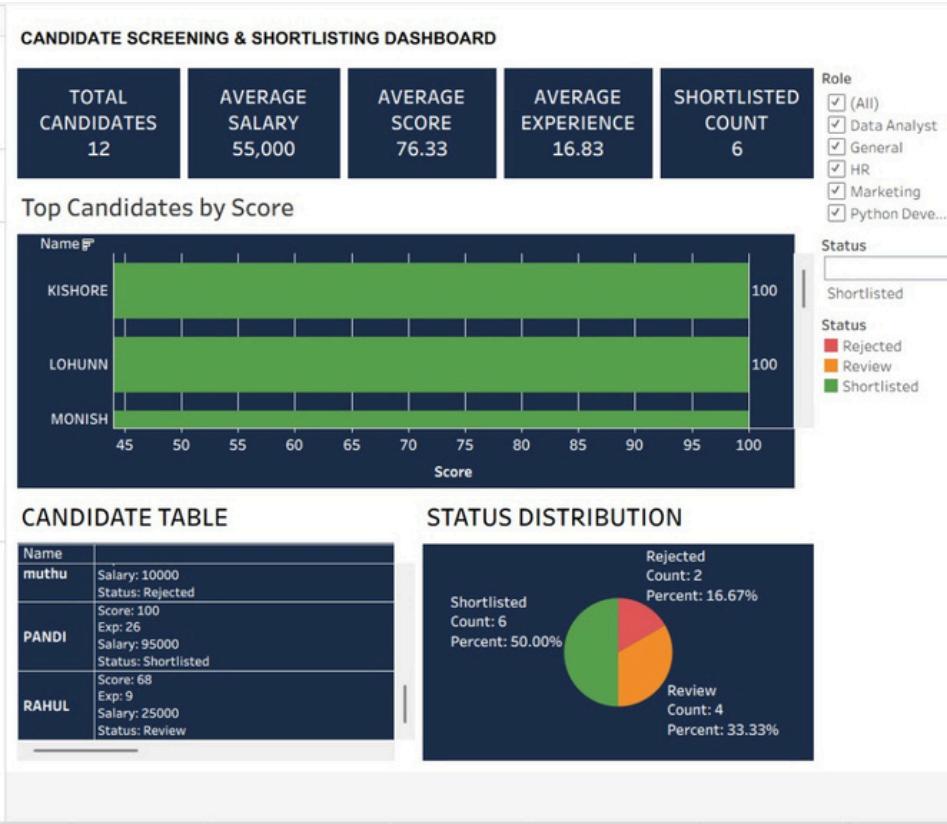
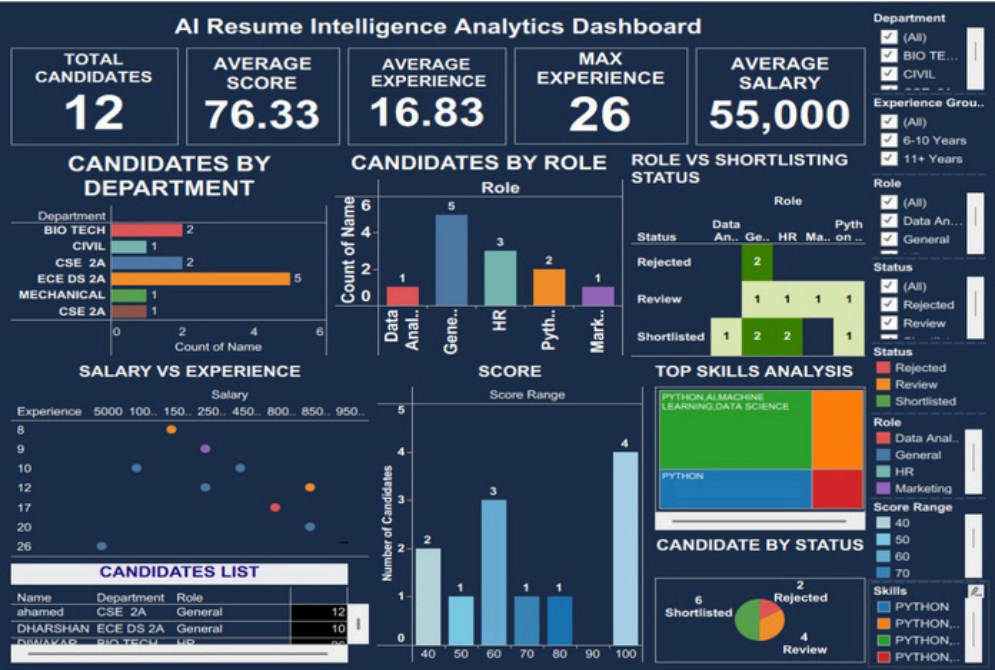


Tableau Dashboard



Advantages

The AI Resume Intelligence and Recruitment Analytics System provides several advantages compared to traditional recruitment systems.

Advantages of the System

- Reduces manual recruitment workload
- Automates candidate screening and evaluation
- Improves recruitment speed and efficiency
- Provides centralized candidate management
- Supports AI-based resume scoring
- Generates professional PDF reports automatically
- Enables QR-based candidate registration
- Provides interactive recruitment dashboards
- Supports real-time analytics visualization
- Improves recruitment decision-making
- Reduces human errors in recruitment operations
- Supports structured database management using MySQL
- Provides better recruitment tracking and monitoring
- Simplifies data import and export operations

The system combines automation, analytics, and dashboard visualization into a single intelligent recruitment platform.

Future Scope

The project can be enhanced further by integrating advanced AI technologies and cloud-based systems.

Future Enhancements

- NLP-based resume understanding and analysis
- Advanced AI candidate ranking system
- Cloud database integration
- Online recruitment portal integration

- Real-time dashboard auto-refresh
- Email notification system for recruiters and candidates
- Online interview scheduling system
- Mobile application development
- Deep learning-based candidate prediction
- Cloud dashboard hosting
- Voice-assisted recruitment system
- Real-time collaborative recruitment management

Future improvements can increase scalability, automation capabilities, and analytical intelligence of the system.

Conclusion

The AI Resume Intelligence and Recruitment Analytics System successfully integrates recruitment automation, AI-based resume evaluation, dashboard analytics, database management, and report generation into a centralized intelligent recruitment platform.

The project simplifies recruitment operations by automating candidate screening, data management, analytics generation, and reporting processes. The integration of Python automation, MySQL database systems, Tableau dashboards, QR-based candidate entry, and PDF report generation improved recruitment efficiency and analytical capabilities.

The system provides:

- Faster candidate screening
- Intelligent recruitment analytics
- Centralized data management
- Interactive dashboard visualization
- Automated report generation

Compared to traditional recruitment systems, the proposed platform offers better scalability, automation, and business intelligence support.

This project significantly improved our practical knowledge in:

- Python programming
- Database integration
- Dashboard analytics
- Machine learning
- Recruitment automation systems
- Business intelligence visualization

Overall, the project provides an efficient and scalable solution for modern intelligent recruitment management and analytics.

References

1. Python Official Documentation
2. MySQL Official Documentation
3. Flask Official Documentation
4. Tableau Documentation
5. Scikit-learn Documentation
6. Pandas Documentation
7. Matplotlib Documentation
8. ReportLab Documentation
9. pdfplumber Documentation
10. python-docx Documentation
11. Research papers on AI-based Recruitment Systems
12. Research papers on Dashboard Analytics and Business Intelligence Systems

Thank You